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DEVELOPMENT ARTICLE

# Adaptive scaffolding and engagement in digital game-based learning

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**Abstract**
Educational games are becoming increasingly prevalent. Recently, adaptive game-based learning to accommodate diverse learners has received considerable attention. The current study aims to explore the effect of adaptive scaffolding from a multidimensional engagement perspective. A total of 61 students from a Taiwan secondary school studied Newton's laws in a computer-based interactive game environment. The students were assigned to either a fixed scaffolding group or an adaptive scaffolding group to test the effect of adaptive scaffolding. The findings suggest that adaptive scaffolding can have a significant effect on students' learning performance and engagement. The study offers significant implications for the design and integration of adaptive scaffold to promote learning engagement in digital game-based learning.

**Keywords** Digital games · Game-based learning · Adaptive scaffolding · Engagement · Learning performance · Science learning

## Introduction

Digital game-based learning (DGBL) has become increasingly prevalent in the educational context. Although DGBL is promising in engaging learners and improving learning, current meta-analyses and systematic reviews have reported inconsistent empirical evidence (Clark et al., 2016a, b; Girard et al., 2013; National Research Council, 2011; Sitzmann, 2011; Vogel et al., 2006, Wouters et al., 2013; Young et al., 2012). Challenges remain in effectively utilizing DGBL to promote learning, especially in the subject areas of science and math (National Research Council, 2011; Young et al., 2012). To help learners connect the game world with science knowledge, providing appropriate scaffolding is essential (Wouters et al., 2008).

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Scaffolding has been applied in various domains to enhance learning or model the steps in the task that would be beyond what the learner could accomplish independently (Vygotsky, 1978). Adaptive scaffolding has shown promises in promoting learning (Anderson et al., 1985; Azevedo et al., 2004; Lim et al., 2023), but few studies have implemented and investigated adaptive scaffolding in DGBL. Concerns exist regarding the possibility of scaffolds interrupting game flow and learner autonomy (Ke, 2009; Leutner, 1993). Furthermore, there is a lack of research investigating the impact of adaptive scaffolding on learners' engagement in DGBL, and the existing studies have not examined engagement as a multidimensional construct (Fredrick et al., 2016). To bridge the research gaps and establish empirical evidence regarding DGBL's benefits for learning, we developed and investigated adaptive scaffolding in a digital game for science learning. In comparing students who played the game with the support of adaptive scaffolding with those who were supported with fixed scaffolding, we were interested in examining both the learning outcomes and students' behavioral, cognitive, and emotional engagement.

## Review of relevant literature

### Digital game-based learning

Digital game-based learning (DGBL) involves several distinct elements such as fantasy, stories, curiosity, feedback, competition, modality, and interaction (Ke, 2016). Through interacting with these features, players achieve learning objectives intentionally embedded in the gameplay (Wouters et al., 2013). Researchers have claimed that DGBL is not only informative but also entertaining, which may contribute significantly to various aspects of learning including cognitive, affective, or behavioral processes and outcomes (Clark et al., 2016a, 2016b; Connolly et al., 2012; Lamb et al., 2018; Vlachopoulos & Makri, 2017; Vogel et al., 2006). Empirical evidence for the claim remains unclear as some meta-analyses and systematic reviews suggested positive effects of DGBL (Clark et al., 2016a, 2016b; Sitzmann, 2011; Vogel et al., 2006; Wouters et al., 2013) whereas others showed less promising results, especially in the subject areas of math and science (Girard et al., 2012; National Research Council, 2011; Young et al., 2012).

One challenge associated with DGBL in science is the connection between tacit knowledge in the concrete game world and explicit, yet often abstract, knowledge in science (Barzilai & Blau, 2014; Wouters et al., 2013). To help players construct science knowledge while engaging with the game world, explicit instructional support, or scaffolding, is often necessary (Wouters et al., 2008). The next section provides an overview of scaffolding and its use in DGBL.

### Scaffolding in DGBL

The notion of scaffolding was introduced by Wood et al. (1976) who defined it as the assistance provided to a child or novice to accomplish a task or goal beyond the individual's unassisted effort. Scaffolding can be provided by humans (tutors or peers), technology (e.g., intelligent tutoring systems), or a combination of both (Ge et al., 2012). While taking on different forms (e.g., question prompts, modeling, process display) and serving different purposes (e.g., cognitive, metacognitive, emotional), scaffolds can be conceptualized into two broad types: *hard* scaffolds provide fixed and static support that is not adaptive to

individual learners whereas *soft* scaffolds adapt to learners' dynamic needs (Saye & Brush, 2002). Several meta-analyses and systematic reviews have reported the positive impact of scaffolding in a variety of contexts from online information search (Zhou & Lam, 2019) to STEM education (Belland et al., 2017; Devolder et al., 2012).

In DGBL, researchers have accentuated the importance of scaffolding to augment the learning experience in the gameplay (Kirschner et al., 2006; Wouters & van Oostendorp, 2013). Ke (2016) operationally characterized DGBL scaffolds based on core game-design elements such as cues, content, feedback, guiding, prompts, or structures. Scaffolding plays a key role in directing players' attention to important information that is often overlooked during gameplay, helping them to organize and integrate knowledge while building connections between game content and disciplinary content (Charsky & Mims, 2008; Feng & Chen, 2014; Sun et al., 2011). Researchers have explored the effects of different scaffolding interventions in DGBL. For example, a skilled in-game agent who prompted students to explain their strategies in a 2D math game was found to increase math performance (Pareto et al., 2012). Similarly, O'Neil et al. (2014) found that self-explanation prompts in a math puzzle game helped students to connect abstract and concrete concepts which had a significant impact on learning performance. In another study, a concept map was embedded in a role-playing game to improve elementary students' performance in natural science courses (Hwang et al., 2013). More recently, Chen, Huang, & Liu (2020) used prompts to scaffold secondary students' science inquiries in DGBL, which led to significant improvement in conceptual understanding in physics; game behavioral data further revealed that the scaffolds helped the students to maintain a balance between playing the game and engaging in the subject matter in the game. Overall, the literature appears to suggest the essential role and benefits of scaffolding in DGBL (Wouters & van Oostendorp, 2013).

On the other hand, DGBL generally employs hard, rather than soft, scaffolds to support learning, although the theoretical foundation of scaffolding entails the concept of adaptivity (Ge et al., 2012; Vygotsky, 1978). Hard or fixed scaffolds have several drawbacks: they could be intrusive, imposing a high cognitive load, affecting the flow of learners' game engagement, and consequently diminishing learner motivation (Adams & Clark, 2014; Charsky & Ressler, 2011; O'Neil et al., 2014; Schrader & Bastiaens 2012). Adaptive scaffolding has a relatively long history in non-game research including intelligent tutoring systems (Anderson et al., 1985), on-demand scaffolds provided by human tutors (Azevedo et al., 2004), and more recently real-time adaptive scaffolds facilitated by artificial intelligence (Lim et al., 2023). In the context of DGBL, a few attempts have been made to offer adaptive advice (Leemkuil & de Jong, 2012), adaptive difficulty levels (Ronimus et al., 2014), or adaptive feedback (Hooshyar et al., 2021), but DGBL research is generally lacking in adaptive scaffolding. Therefore, this study sought to investigate adaptive scaffolding in DGBL. In doing so, we were interested in examining not only the learning outcomes but also learners' engagement in the process. The next section introduces the current study.

## Adaptive scaffolds and engagement in DGBL

In the current study, we aimed to provide and investigate adaptive scaffolds to support science learning in DGBL. Specifically, the game system will analyze students' interactions with different game features and provide adaptive support to help the students succeed in the game and achieve target learning objectives.

In addition to examining the effects of adaptive scaffolds on science learning outcomes, we were also interested in learners' engagement, an important factor that has received

much attention in DGBL and educational research in general (Fredricks et al., 2004, 2016) distinguished three components of engagement: *cognitive* engagement is the mental effort learners invest in learning and their self-regulation processes (Finn & Zimmer, 2012; Greene, 2015), *emotional* engagement refers to learners' positive or negative emotional reactions such as enjoyment, fun, excitement, or boredom (Filsecker & Kerres, 2014; Jung & Lee, 2018), and *behavioral* engagement is related to learner behaviors such as participation, attention, effort (Fredricks et al., 2004; Goggins & Xing, 2016). In regard to cognitive engagement, Miller et al. (1996) further delineated two dimensions: self-regulation and deep strategy.

Engagement has been studied in the DGBL context. For example, Pellas (2014) found that cognitive, emotional, and behavioral engagement were correlated with each other in DGBL. Presumably, games have the potential to engage learners, but research has not reached a consensus regarding the effect. A recent systematic review suggested that certain game design features can improve learner engagement (Abdul Jabbar & Felicia, 2015). For adaptive scaffolds in particular, a recent meta-analysis by Liu et al. (2020) did not find evidence that the scaffolds enhanced learner engagement in games. Liu and colleagues were only able to locate four empirical studies that examined engagement as a dependent variable by comparing adaptive scaffolds with a control condition. It is worth nothing that none of the four studies adopted a comprehensive framework of engagement by examining its multiple dimensions. Specifically, Clark et al. (2016a, 2016b) used a 4-item short survey to gauge students' level of engagement, Long and Aleven (2014) measured enjoyment as a facet of emotional engagement, whereas the other two studies examined some types of behavioral engagement (Kuk et al., 2017; Shabihi et al., 2016). Since engagement is a multidimensional construct (Filsecker & Kerrs, 2014; Martey et al., 2014), we argue that it is important to explore different dimensions of engagement to gain a better understand of the impact of adaptive scaffolding. Therefore, this study adopted the frameworks by Fredrick et al. (2016) and Miller et al. (1996) to examine cognitive, emotional, and behavioral engagement in DGBL. Further, considering the relationship between engagement and learners' in-game performance as well as learning outcomes (Eseryel et al., 2014; Chen et al., 2019; Ding et al., 2017; Sabourin & Lester, 2014), we postulate that learners' engagement in DGBL may be a moderating factor in the relationship between scaffolding and learning outcomes. Taken together, this study set out to investigate the following questions in the context of DGBL:

1. Do students supported with adaptive scaffolding perform better than students with fixed scaffolding?
2. Do students supported with adaptive scaffolding show more engagement than students with fixed scaffolding?
3. Does students' engagement moderate the relationship between the types of scaffolding and their learning performance?

## Method

### Design and participants

To answer the research questions, we adopted the quasi-experimental design approach. Sixty-one 7th grade students (mean age 13) from a secondary school in Taiwan voluntarily

participated in the study. Students were randomly assigned by class to two groups: a fixed scaffold group (N = 30; 17 males and 13 females) and an adaptive scaffold group (N = 31; 16 males and 15 females). The students were from the same school district, sharing similar socio-economic status. Among the participating students, 90% reported experience playing non-educational games, but only 22% had experience with educational games.

## The game environment: summon of magicrystal

In this study, students played a game named *Summon of Magicrystal*. *Summon of Magicrystal* is a computer-based interactive game focusing on Newton's laws, and the target learning objectives were aligned with secondary curriculum standards. The game portrayed a village that was under attack by a hoarding dragon, and players were charged with saving the village by securing its food supply in a timely manner. In addressing various challenges or problems in their rescue process, players interacted with different objects or characters in the game world where physics concepts such as time, distance, position, and velocity were operationalized in the interaction. For example, students could use speed control to adjust the transportation speed of certain food and see results or use the timer tool to measure the time taken for certain food to arrive. Students' interactions with the different game elements (e.g., timer, speed control) were tracked and recorded by the game database. Throughout the process, students accumulated points, advanced levels, and ultimately accomplished the mission while simultaneously establishing an understanding of the physics concepts and principles, thereby achieving intended learning objectives.

## Treatment: fixed vs. adaptive scaffolds

While playing *Summon of Magicrystal*, the two groups of students received support in different ways. At each challenge level, the students in the *fixed* scaffolding group received scaffolds that explained relevant concepts, clarified a task, or prompted them to further consider a problem. As illustrated in Fig. 1, to help students solve problems using path length and speed according to the movement of the objects, the system would present a question for students to answer. Additional hints are also provided to students. Students in this group received all the pre-planned scaffolds in a fixed sequence. The scaffolds were developed by two experts who were not only experienced in the science content but also specialized in the game environment.

For the *adaptive* scaffolding group, the same set of scaffolds was available in the game system, but a scaffold may or may not be triggered to appear, depending on the system's diagnosis of an individual student's status. As shown in Fig. 2, the automated adaptive scaffolding (AAS) model involves the following processes: (1) The game system collects students' log data, game progress, and answer results, and records them in the database. (2) The system analyzes a student's status based on the data, and these data are used for scaffolding auxiliary systems, capability indicators, or behavioral analysis and statistics. (3) The game analyzes the level of students and possible obstacles to evaluate or describe students' familiarity with the learning content. (4) For students who are evaluated or portrayed through ability indicators and scaffolding assistance systems, the game continues to collect their behaviors until certain specific behaviors reach the expected number of times or conditions, indicating what the student needs. Students with different cognitive levels have different conditions that trigger them to different topics to assist them at various cognitive levels. (5) The scaffolding assistance in the game provides the content of the

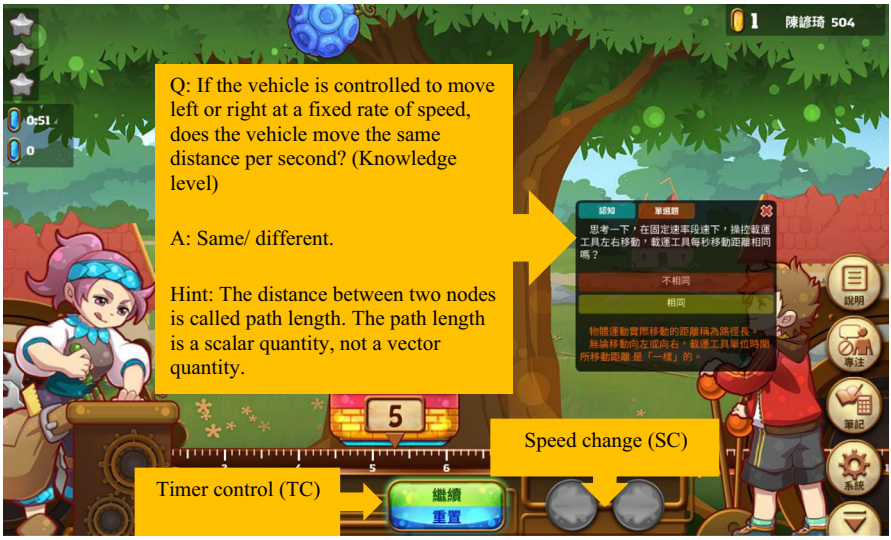


**Fig. 1** An example scaffold in the game

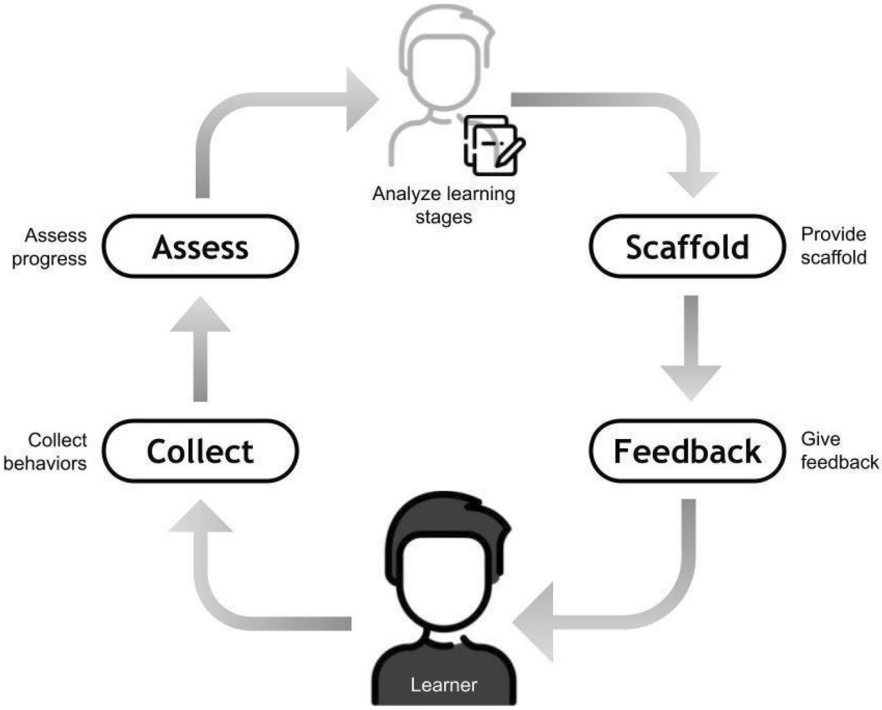


**Fig. 2** The automated adaptive scaffolding (AAS) model in the game

scaffolds corresponding individual students' cognitive levels and encountered obstacles, so as to enhance their cognition or eliminate any conceptual misconceptions. (6) The scaffolding auxiliary system has a fault-tolerant mechanism and a feedback mechanism that can give answers and explanations to ensure that students can internalize their own cognition from the process.

As explained in Fig. 2, the provision of scaffolds was based on a pre-defined coding framework (Chen, 2019). The system identifies an individual student's behavior data (e.g., frequency of using the time control), compare it with the calculated means of all students' behavioral data stored in the game system (e.g., the mean frequency of all students' use of the time control), and determine the distance of the student's behavior data from the mean data, which is measured in standard deviation units. More specifically, we first calculated the $z$-score (Eq. 1), where x is the raw score, μ is the population mean, and σ is the population standard deviation.

$$z = \frac{(x - \mu)}{\sigma} \tag{1}$$

Next, we use Eq. 2 to calculate the proportion of the specific behavior's z-score in the sum (Eq. 2)

$$\frac{\sum specific\ behavior}{\sum z.score} \tag{2}$$

Finally, the model uses test value 1 as the activation function to output the probability vector. To establish the parameter optimizer, the upper limit for triggering was set at 15 behavior frequencies (Eq. 3). The result is rounded to the nearest whole number.

$$15 \times (1 - \frac{\sum specific\ behavior}{\sum z.score}) \tag{3}$$

To illustrate with an example, the scaffold shown in Fig. 2 would not appear to the *adaptive* scaffolding group unless their behaviors triggered it. For this particular example, when a student's frequency of using the timer control behavior is stored seven times (Eq. 3), then the scaffold would be triggered and provided to the student.

$$15 \times \left(1 - \frac{38.58}{68.65}\right) = 6.57028$$

## Measures

The pre-test and post-test included the same 30 items to measure students' conceptual understanding of the physics content covered in the game. In addition, the measurement of students' engagement was adapted from validated instruments (Fredricks et al., 2005; Miller et al., 1996), including four subscales: behavioral engagement, emotional engagement, self-regulation, and deep strategy, with the latter two representing cognitive engagement. To ensure strong internal validity, two items from the behavioral engagement scale and one item from the emotional engagement scale were removed so that Cronbach's alphas of all scales were above 0.70 (Hair et al., 2006). Example items and reliability coefficients are presented in Table 1.

**Table 1** Example items and reliability coefficients of the engagement scales

| Subscales | # of items | Example Item | *a* |
|---|---|---|---|
| Self-Regulation | 8 | When I finish working a problem, I check my answer to see if it is reasonable. | 0.90 |
| Deep Strategy | 5 | I draw pictures or diagrams to help me solve some problems. | 0.89 |
| Behavioral engagement | 4 | If I have trouble understanding a problem, I go over it again until I understand it. | 0.75 |
| Emotional engagement | 3 | I am very interested in learning. | 0.86 |

### Data analysis

To answer the first research question regarding the effect of scaffolding on performance and third research questions regarding the moderating role of engagement, a 2×2 factorial ANCOVA was performed on students' performance (post-test scores), with pre-test scores as a covariate. The first factor was the type of scaffolding (adaptive vs. fixed), and the second factor was the type of engagement (self-regulation, deep strategy, behavioral, and emotional). We used mean scores to split the data for each engagement scale into two groups – high engagement and low engagement. To answer the second research question, we ran one-way MANOVA to test the effect of scaffolding on the four types of engagement. Levene's tests for the ANOVAs showed insignificant results ($p > .05$), suggesting that the homogeneity of variance assumption was met.

### Procedure

This study took place in the computer classroom. Participating students were first informed about the purpose and procedure of the study. Then, the students took the pre-test that measured their conceptual understanding. Subsequently, the students were randomly assigned by class to either fixed or adaptive conditions, and were given username and password to log in the game environment. Students worked individually on their own computer. The duration of the study was about six hours lasting two weeks. Upon their completion of the gameplay, students took the post-test and follow-up engagement questionnaires.

## Results

Descriptive statistics of the key variables (conceptual tests and engagement scales) are presented in Table 2. Correlations among the variables are reported in Table 3.

To answer the first research question regarding the group difference in the learning performance, the results of ANCOVA indicated that adaptive scaffold had a significant effect on students' performance, $F(1,59) = 17.847$, $p = .00$, $\eta^2 = 0.263$. Students with adaptive scaffolds performed better than students with fixed scaffolds with a large effect size.

To answer the second research question regarding the group difference in students' engagement, the MANOVA results on the four engagement scales suggested that the

**Table 2** Means and standard deviations (SD) across groups

| Variable | Fixed Scaffolding | | Adaptive Scaffolding | |
|---|---|---|---|---|
| | Mean | SD | Mean | SD |
| 1. Pre-test | 13.07 | 3.99 | 13.77 | 3.90 |
| 2. Post-test | 14.93 | 4.73 | 19.19 | 4.05 |
| 3. Self-regulation | 2.96 | 0.79 | 3.49 | 0.90 |
| 4. Deep strategy | 2.73 | 0.90 | 3.38 | 0.95 |
| 5. Behavioral engagement | 3.31 | 0.87 | 3.35 | 1.00 |
| 6. Emotional engagement | 2.63 | 0.50 | 3.03 | 0.71 |

$N_{Fixed-Scaffold} = 30$; $N_{Adaptive-Scaffold} = 31$

**Table 3** Correlations among variables

| Variable | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1. Pre-test | 1.00 | 0.58** | 0.23 | 0.24 | 0.41** | 0.18 |
| 2. Post-test | | 1.00 | 0.42** | 0.36** | 0.41** | 0.34** |
| 3. Self-regulation | | | 1.00 | 0.82** | 0.48** | 0.46** |
| 4. Deep Strategy | | | | 1.00 | 0.42** | 0.47** |
| 5. Behavioral Engagement | | | | | 1.00 | 0.38** |
| 6. Emotional Engagement | | | | | | 1.00 |

adaptive scaffolding group engaged better than the fixed scaffolding group (*Wilks'* $\Lambda = 0.815$; $p = .02$). Specifically, students with adaptive scaffolding showed better engagement in three aspects: self-regulation, $F(1,59) = 6.008$, $p = .017$, $\eta^2 = 0.092$, deep strategy, $F(1,59) = 7.611$, $p = .008$, $\eta^2 = 0.114$, and emotional engagement, $F(1,59) = 6.586$, $p = .013$, $\eta^2 = 0.10$. Nevertheless, the two groups did not have a significant difference in behavioral engagement, $F(1,59) = 0.026$, $p = .873$, $\eta^2 = 0.000$.

To answer the third research question regarding the moderating role of engagement in the relationship between scaffolding and learning performance, we performed ANCOVA to test the interaction of scaffolding types and engagement on learning performance. The results suggested that none of the four types of engagement moderated the effect between scaffolding types and student performance ($p > .05$).

## Discussion

Adaptive learning has received considerable attention in recent years due to its potential to support learners with diverse backgrounds (Shute & Zapata-Rivera, 2012; Lim et al., 2023). Yet in DGBL, learning is often supported with fixed scaffolds. There is a lack of empirical research on adaptive scaffolding in DGBL (Liu et al., 2020). This study set out to investigate adaptive scaffolding in DGBL from an engagement perspective. Overall, the findings show promises of adaptive scaffolding in DGBL.

Compared with fixed scaffolds, the adaptive scaffolds deployed in the DGBL environment of this study helped students in the adaptive group to develop a significantly better conceptual understanding of physics. The finding is promising, given the mixed results reported in the literature regarding the effect of DGBL on science learning (Girard et al., 2013; National Research Council, 2011; Young et al., 2012). The finding joins those from non-game research to suggest the benefit of adaptive scaffolds over fixed scaffolds in supporting student learning (Azevedo et al., 2004; Leemkuil & de Jong, 2012; Liu et al., 2020; Shute et al., 2012). Further, while adaptive scaffolding has been implemented in the existing few DGBL studies as adaptive advice, difficulty levels, or feedback (Leemkuil & de Jong, 2012; Hooshyar et al., 2021; Ronimus et al., 2014), this study is among the first attempts to generate adaptive scaffolds based on a systemic analysis of learners' tool access, game progress, and answer results in the gameplay.

What caused the significantly better performance of the adaptive scaffolding group? The multidimensional engagement framework adopted in this study (Fredricks et al., 2004; Miller et al., 1996) afforded us insights into the question. Among the four scales measuring students' cognitive, behavioral, and emotional engagement, the adaptive scaffolding groups showed significantly more engagement in three scales representing cognitive and emotional engagement. It appears that the adaptive scaffolds offered in gameplay prompted the students to consciously regulate their learning and employ deep strategies in the process, which plausibly, led to their superior performance. In addition, the students' positive emotional engagement might also have contributed to their performance. Interestingly, the two groups of students did not differ in their behavioral engagement. We postulate that the reason might lie in the very nature of adaptive scaffolding. While the students in the fixed scaffolding group had to address all the pre-planned scaffolds regardless of their mastery, the students in the adaptive group only focused on those targeting their levels. As such, the adaptive group might have been more deeply engaged cognitive-wise, yet the students did not necessarily show a higher level of participation, attention, persistence, or effort. Comparatively, the demand to address all scaffolding prompts might have imposed a high cognitive load on the fixed scaffolding group. As a result, students in this group were not able to focus on their unique learning needs, which might also have a negative impact on the students' emotional engagement.

Although the value of DGBL to support learning engagement has been well established (e.g., Abdul Jabbar & Felicia 2015), few studies have investigated how adaptive scaffolding promotes student engagement in DGBL, and the existing few attempts show inclusive evidence (Liu et al., 2020). Our study added some clarity to the effects of adaptive DGBL on engagement and further reinforced the need to investigate engagement in DGBL as a complex and multidimensional construct (Filsecker & Kerrs, 2014).

Finally, we tried to explore the moderating role of engagement in the relationship between types of scaffolding and learning performance, but our findings did not find a significant interaction effect between engagement and types of scaffolding. Thus, while engagement has important bearings on learning outcomes (Eseryel et al., 2014; Chen et al., 2019; Ding et al., 2017; Sabourin & Lester, 2014), their impact did not vary with types of scaffolding provided in DGBL.

## Conclusions and implications

This study provided solid empirical evidence for the benefit of adaptive scaffolding in supporting both science learning performance and engagement in the DGBL environment. It sheds light on future directions of using DGBL to support science learning, especially given the currently ambiguous findings in the literature. This study shows promise of developing adaptive scaffolds based on learners' log data and in-game progress and performance. Future research should further investigate various approaches to developing adaptive scaffolding in DGBL. Further, adaptive scaffolds have the potential to balance play and learning without overwhelming learners and diminishing their motivation. Future research should further investigate the effects of adaptive scaffolding on learner motivation. In addition, to better assess emotional engagement in DGBL, future researchers could consider a more thorough and inclusive measurement.

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## Declarations

**Conflict of interest** The authors declare that they have no conflict of interest. Access to the original dataused in the present study in anonymized form is granted by an application to thestudy's first author. This study involved only voluntary participation. We obtainedinformed consent from all the participants, who were also informed that they couldstop participating at any time and that whether or not they participated in the studywould not affect their course grades. All the information collected from this study iskept confidential.

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